

Android mobile application dedicated to inform about health and different diseases that are most common to humans

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I. INTRODUCTION

The paper is based on the fact that nowadays, most of the activities, which any person carries out, can be related to a mobile phone or a personal computer. Based on this statement, a study found on the internet ¹ shows that most people prefer to use a mobile phone to make their daily tasks easier, such as scheduling, shopping or paying bills, with a few touches. Another survey found online [1], showed that the preference for Android applications is higher than for a similar web application, and the main argument is that this type of development can run on any device that uses the Android operating system, but also on the Windows operating system that has an emulator.

Currently, most activities have been moved online or by phone, in order to avoid as much as possible, the physical interaction between people, but there are areas where the activity cannot be moved completely or even partially in a virtual form, and their activity has been reduced to a minimum. In this paper will be implemented an application that helps people who due to the current situation cannot get in direct contact with a doctor, because after a telephone consultation were not introduced in the category of people in urgent need of physical consultation.

The subject that generated this idea was related to the possibility of every person to get acknowledge and explanations about different symptoms that may occur in different parts of the world. The low medical staff in different regions and the old types of

practicing medicine should be a sign of people asking for help.

Different studies that can be found online show the fact that a lot of children under the age of five are losing their life because of disease like respiratory infections, diabetes, nutritional deficiencies or digestive diseases. The number of losts are bigger in different parts of Africa and Asia where the states are less developed [2].

The mobile application was the best alternative to develop this project, in my opinion, because of the expenses that can appear at the purchase of a mobile phone in comparison with a personal computer. After looking on different applications marketplace I realized that I cannot find an app that can ask all my questions when I am sick, like *how bad is it? where can I find medical personnel specialize in my affection? can one treatment that I am on affect my current state? what is the recommended dose of my treatment?*

Most of these questions can be answered only by a doctor, but in some situations a little help can save us time and money spent at medical offices. I am aware of the fact that this type of informative material cannot replace medical instructed personnel.

The projects wants to help informing people about health, that is why the functions that can be found in this development are setting alarms for treatments, section dedicated to different diseases whit informations about symptoms and possible evolutions without treatment, in multiple languages. Another interface of this application is dedicated to a contact page detailed with the schedule and

cabinets addresses of doctors that have different specialties.

Beside the functionalities already mentioned before, there will be the possibility of scanning a treatment bar code to see it's prospect.

One last property, that was quite interesting to me when I done the research for this work, was the possibility of implementing a question-based generator. The purpose of this is to generate a possible diagnostic based on the answers given by the user to the questions that are stored in the application data base. Every answer is the key to the next question that will be generated. This method of diagnostician will recommend what type of specialist the user needs to see.

My approach tries to offer a contact to useful information for all the people that have in possession an android system base mobile or an personal computer with an android emulator. The scope of this implementation is to grow the common knowledge of everyone about essential symptoms, disease or treatments. A better understanding of the human anatomy and it's immunity system, along with indications and treatments given by doctors, using virtual platforms can offer advantages. Most of the datas that will be use in this project will be local, mostly specific for our country, because a worldwide coverage would need a larger period of time and a bigger development team.

II. STATE OF THE ART

Nowadays we can see what a medical crisis means, what rules and restrictions can appear, and which are the consequences of our behavior. During different periods in history, humanity confronted with pandemics like Spanish flu, tuberculosis, black plague and now Covid-19. Because of the period in time those occurred, the medical system was in different states of evolution and the methods used had a certain level of success and different ways of treating the patients.

In the research part of my activity, I was focused to find different types of applications that can be considered as a part of healthcare category. After this

I tried to make a separation between the fields those apps were focused on. In the applications marketplace under the category healthcare can be found apps dedicated to nutrition, diabetes, sports or human body anatomy among others.

A. Applications focused on diagnosing

In this section of the second chapter the attention is going to the importance of a good information and an early diagnostication.

In this area of interest, were developed applications like *EasyDetectDisease* and *Ada-check your health*.

EasyDetectDisease was implemented starting from the idea that a lot of parents from Pakistan can not identify the problem that affect their childrens only by seeing some symptoms. Before starting the code writing, the developers have made a research to see the percent of low educated parents. This idea helped them to establish the way of presenting the information into the application, so it can be used by everyone. The level of study in the country determine the programmers to make the application very intuitive, so it can be used by those people that have no high education or very expensive mobile phones. The main way of information presentation, in the application, is visual by using videos and photographs. Those were used to explain different methods of treatment or preventions, or to give an example of symptoms that can appear at different disease that can affect the kids. All the information that was used in the *EasyDetectDisease* was verified by medical staff before being introduced as information in the app. The study that was done to determine the rate of deaths and the infections among kids during the years shown that the cases are decreasing, but there is still that category of families that can not afford an medical intervention for every type of infections that can seem not that dangerous from our point of view.

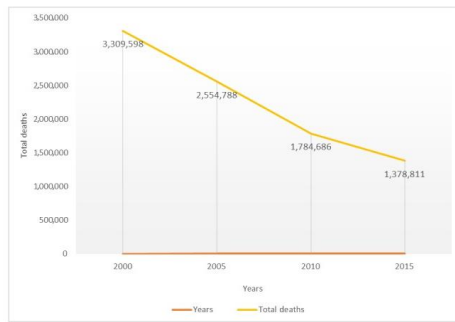


Fig. 1-Graphic representation of the number of death between 2000-2015 among childrens under five years old

In the table can be seen the number of children under five years old that died during a period of fifteen years because of infectious and parasitic diseases in Pakistan. With all of the above being said, EasyDetectDisease is an application dedicated to the parents in Pakistan from all society layers, being they high or low educated, rich or poor, having the purpose of helping them to treat or prevent diseases like digestive or respiratory ones before any complications appear.

Another application mentioned at the beginning of this section is Ada-check your health, which is actually more like a chatbot that can help all peoples to reach a diagnostic. This one is very easy to understand because the idea that can be found at the basis of this app is the following one: a person is asked about the way he/she is feeling and based on the answer the application is selecting the next one until the point when an diagnostic can be predicted. Ada can be seen like an virtual medical attendant that can offer an opinion about what disease we have, but at the end of every identification a specialized opinion of a doctor is recommended.

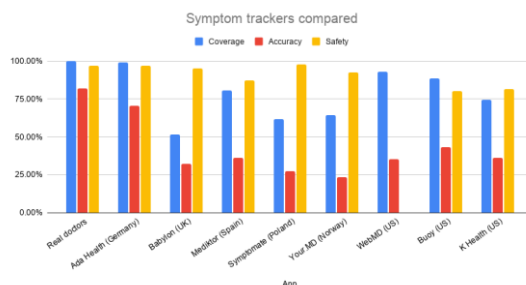


Fig. 2-Statistics about Ada's application accuracy in different countries

In the table can be seen three factors of interest when it comes about the comparison between Ada and real doctors, accuracy, safety and coverage; the study was done in different countries. Besides this two applications can be found many more, but those are similar, when we talk about functionalities, with this ones.

B. Applications used to scan bar codes

The second type of relevant android implementation is the one that involves the development of a barcode scanning functionality. This possibility is based on the desire to have the most relevant information, or rather, the most important (such as dosage, side effects, ingredients) for different categories of medication, just a touch away at any time of the day or in any location, without the user of the application having to search for the prospectus.

In this section of the chapter I will present an application, ShopEZY, that is used to make shopping more easy. The principle of operation of the ShopEZY application is as follows: a user of the application uses ShopEZY, in a store, to scan the barcode of the products, when all the details of the product are displayed on the screen of the mobile phone, they are to be sent to the store server using wi-fi connections. Each scanned product goes through this process, then is added to the list of scanned products formed in the application. The final step involves transforming the list, formed within the application, into a tax receipt and sending it to the store system, so that then the user can pay for purchases and leave[3].

C. Applications that can set alarms and check doctors disponibility

This category of applications is relevant for my project because, as I described the functionality of my app in the first chapter, one part is where the user can find a doctor specialized in his/hers condition. And another aspect is that the user will have the posibility to set different type of alarms to help him/her to respect the traitement.

Another relevant example of an application for the chosen theme is the one in which the application

consists of a web implementation, but also an android application. The Healthcare application has the following functionality: the user describes the symptoms in either of the two types of application, following this action the system identifies and selects doctors, this is done based on their specialty, location and availability of each. Following this selection, the user has the option to choose a time interval for an appointment, from a list displayed within the application. From the point of view of the user in the position of the doctor, he can consult the patient's medical history, can add mentions and treatments. Following an established treatment, notifications will be formed in the personal calendar of the patient user and various alarms will be set to remind him of the medication, so a monitoring of the medical condition of the patient user is performed. Another feature described throughout the document is the reminder for the vaccination period in children, listing the doctors who have availability and the resources needed to administer the dose of vaccine and the possibility to make an appointment with one of the doctors. Eventually the vaccine will be added to the patient's medical history [4].

III. METHODOLOGY

In this chapter I will start to talk about the technologies that I will use in the development of the application, to be more specific, Android Studio and Firebase.

A. Android Studio

Android studio is a development environment (more precisely an integrated development environment, IDE) launched for the first time in 2013 by Google. This development environment collaborates with the Android platform. Android Studio is available to the public in a free version with the release of version 0.1 in 2013. The first stable version was released in the winter of 2014, this was version 1.0. Android studio is the official development environment for android application development. In 2017, Google announced that it wants to add support for a new language used for application development, Kotlin. This IDE is

currently available for Windows, OS X, and Linux [5]. In terms of packages, Android not only uses them to arrange the code in the app, but also to manage the app itself [6].

Any android application developed in Android Studio has a basic structure, without which it would not work properly, which can be seen in the image below.

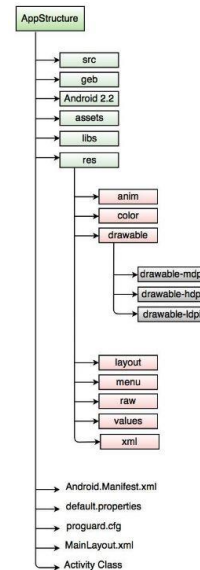


Fig. 3- Android project structure

In order to be able to run the code written in Android Studio on the computer, without loading the program on a device whose operating system is Android, the developer needs an emulator. In order to be able to install an emulator, the following basic requirements must be met (Operating System, Random Access Memory (RAM), Free Disk Space, Minimum Version of JDK, Minimum Screen Resolution) [5]. To see all the files contained by our android project, not only those related to the application module, we must be in Project mode. The meaning of some files in the project folder: .gradle (The file is created every time the project is built and it contains the settings and files needed to build the project, they can be modified or deleted), .idea (this saves the project-specific metadata, it contains a set of configurations each of which is specific to a functional area), app (in this area the code written by

the programmer is saved), build (this folder contains specific data obtained after building the project) [7].

The App folder contains, among others, the following folders: app / build (contains the outputs generated when running the application), app / libs (contains all the libraries used in this module), app / src (here you can find the source code, images, videos, audios, etc. which are used in the module), app / build.gradle (here are included the specific configurations of the module construction) [7]. În fișierul App/src se disting trei foldere mari androidTest (îna cest fișier se scrie cod care efectuează testare care are nevoie de emulator sau de un dispozitiv cu sistem de operare Android), main (acest folder conține secvențele principale din codul sursă necesare construirii proiectului) și test (aici se scrie codul care efectuează testare care nu necesită un emulator sau un dispozitiv android, sunt testele care rulează local pe mașina virtuală. Acest test este utilizat când nu este un framework specific android, cu scopul de a reduce timpul de execuție) [7].

Regarding the structure of the App / src / main file, it contains AndroidManifest.xml (this file is mandatory in every Android project, because it describes the nature of the application and each component of the application. Also here is the name of the java package, used as identifier unique for the application and the minimum level of the API), java (here is the java source code), res (here are the resource files such as strings, colors, sizes, drawable files, layouts, etc.) [7].

B. Firebase

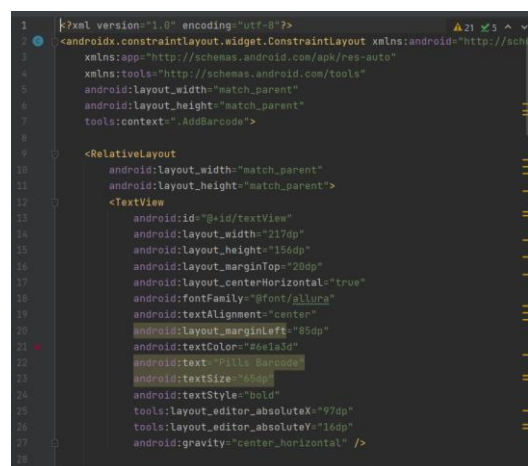
Firebase appeared in 2011, being a project produced by Firebase Inc., this is a platform for the development of mobile and web applications, in 2014 the platform was acquired by Google [8]. Firebase is a platform that provides the user with various tools to provide support for a high quality process in order to facilitate the monitoring of the project, in order to pay high attention to users. Currently Firebase provides the user with 20 products, which are divided into three categories: application development (Cloud Firestore, ML Kit, Cloud Functions, Authentication, Hosting, Cloud

Storage, Realtime Database), improving the quality of the application (Crashlytics, App Distribution, Performance Monitoring, Test Lab) and business growth (Analytics, Extensions, Predictions, Firebase A / B Testing, Cloud Messaging, In-App Messaging, Remote Config, Dynamic Links, App Indexing) [9].

IV. IMPLEMENTATION IDEAS

This chapter is dedicated to the part where I propose a method of implementation for the idea I presented in the last sections.

In Android studio I used two types of files for the code implementation: xml files and java classes. These files represent the base of the application, to be more specific, they are used to define and build an interface of the application. I used the xml files for the design of the activities (interfaces). In these were used components like TextView (used to print a string in the activity), VideoView (used to play a video that is inserted in the application folder at the res file level), Button (for the implementation of the buttons), ImageView (used to add images to the activity), etc. An example of a xml file from the project can be seen in the below picture.



```
1 <?xml version="1.0" encoding="utf-8"?>
2 <androidx.constraintlayout.widget.ConstraintLayout xmlns:android="http://schemas.android.com/apk/res-auto"
3     xmlns:app="http://schemas.android.com/apk/res-auto"
4     xmlns:tools="http://schemas.android.com/tools"
5     android:layout_width="match_parent"
6     android:layout_height="match_parent"
7     tools:context=".AddBarcode">
8
9     <RelativeLayout
10         android:layout_width="match_parent"
11         android:layout_height="match_parent">
12         <TextView
13             android:id="@+id/textView"
14             android:layout_width="217dp"
15             android:layout_height="156dp"
16             android:layout_marginTop="28dp"
17             android:layout_centerHorizontal="true"
18             android:fontFamily="@font/allura"
19             android:textAlignment="center"
20             android:layout_marginLeft="85dp"
21             android:textColor="@color/black"
22             android:text="@string/hello_world"
23             android:textSize="18dp"
24             android:textStyle="bold"
25             tools:layout_editor_absoluteX="97dp"
26             tools:layout_editor_absoluteY="14dp"
27             android:gravity="center_horizontal" />
28
```

Fig. 4- XML file example

Every tool used has some specific parameters that need to be completed, android:layout_width and android:layout_height, with the help of these two can be setted the desired dimension for the used component. With the help of the xml files designs like the one presented below can be created.



Fig. 5- Example of Android application interface created using XML file from the project. This image corresponds to the main page of the application *MedicalFriend* developed by me.

Also the main file of the project is the xml type, and there are defined all the activities created, the style of the entire application and the name of it (AndroidManifest.xml). The structure of this one can be seen in the following picture.

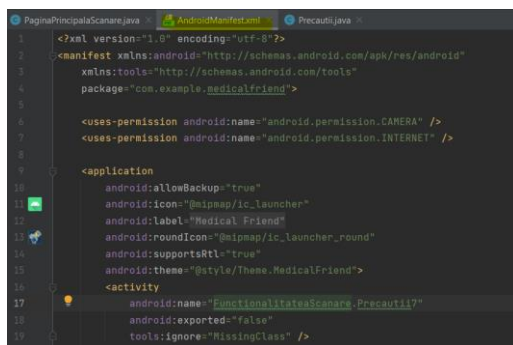


Fig. 6- The AndroidManifest.xml file of the application *MedicalFriend*

Because I implemented the possibility to choose between more languages like Romanian, English, French, and German I needed to create specific xml files to define strings that will be used as statements in the application besides the string.xml file that is created initially. The structure can be seen in the below image.

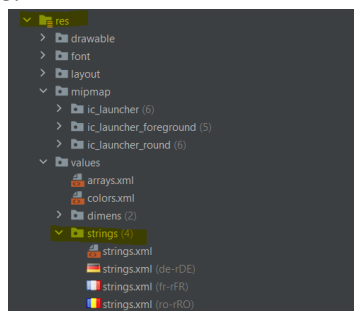


Fig. 7-Structure of the string files used to implement multiple languages in the application

After the design of an activity was done to add the functionality for the buttons or layout I started to write code in the java files. For this reason, in every xml file created for an specific activity, if I want to add a functionality for a specific component I need to give that one an unique id (e.g android:id="@+id/textView") so I can make a link between the design and the function.

For example I used buttons to change between activities, to add text from the autoCompleteTextView in the database (Firebase) or to extract a certain information from the database.

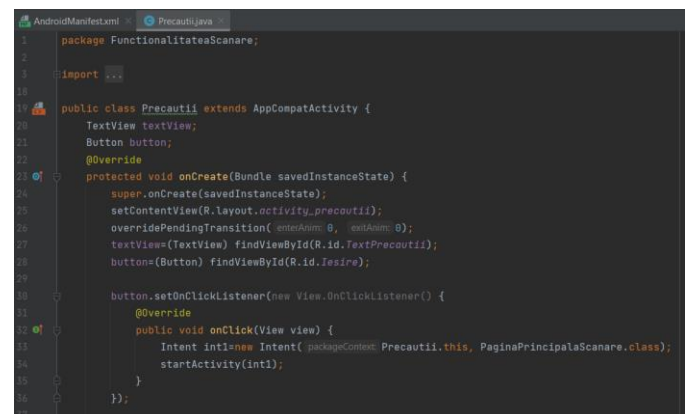


Fig. 8- Java class example for button implementation that helps with the change between the interfaces

Also the connection to the database for insertion, extraction or account creation is done in the Java files. For every tool used that is not default for the java class is added a default class in the import section.

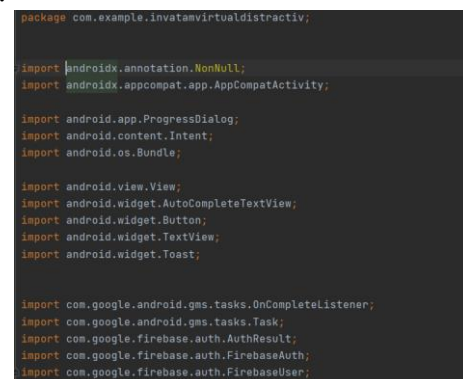


Fig. 9- Example of the import section of a java class

If a tool that I need to used can not be added to the import section that means I have to modify the

build.gradle at the application level to add the implementation for that specific component and to sync the whole project so the modification can be recognized by the application.

```
dependencies {

    implementation 'android.support.test.espresso:espresso-core:3.1.1'
    implementation 'com.google.android.material:material:1.4.0'
    implementation 'androidx.constraintlayout:constraintlayout:2.1.1'
    implementation 'com.google.firebase:firebase-analytics:19.0.2'
    testImplementation 'junit:junit4:1.1.1'
    androidTestImplementation 'androidx.test.ext:junit:1.1.1'
    androidTestImplementation 'androidx.test.espresso:espresso-core:3.4.0'

}

//FIREBASE
// Import the Firebase BoM
implementation platform('com.google.firebase:firebase-bom:28.4.1')
// Add the dependency for the Firebase SDK for Google Analytics
// When using the BoM, don't specify versions in Firebase dependencies
implementation 'com.google.firebase:firebase-analytics'
// Add the dependencies for any other desired Firebase products
// https://firebase.google.com/docs/android/setup#available-libraries
implementation 'com.google.firebase:firebase-analytics:19.0.1'
implementation 'com.google.firebase:firebase-auth:21.0.1'
implementation 'com.google.firebase:firebase-core:21.0.0'
implementation 'com.google.firebase:firebase-clint-android:2.5.2'
implementation 'com.borjabravo.readmoretextview:2.1.0'
implementation 'com.squareup.picasso:picasso:2.71829'
implementation 'com.github.medyo:android-about-page:1.1.1'
```

Fig. 10- Application's build.gradle file

When it comes to the connection between the application and the online database, this is done using both Android Studio and Firebase. Using the Tools menu provided by the development environment, all the needed dependencies are added to the file mentioned before, and by accessing the site of the Firebase from Android Studio will create all the linkage between the android project and the firebase project like package name, will bring locally the json file needed from the database in the project file and the created online project in Firebase console will be the android type, all it remains to do is to give a name to the project and then to activate and create the functionalities that I need to use, from Firebase, in my android project.

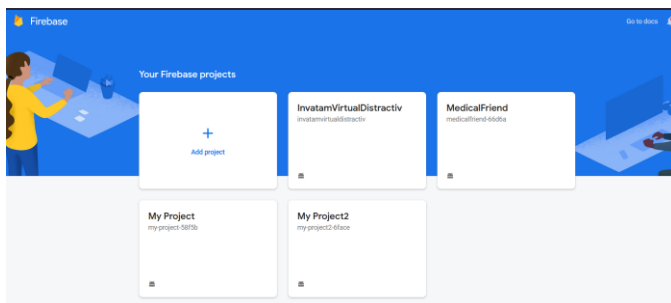


Fig. 11- Firebase Console

At this moment, the project implementation is in the middle, the database linked to the project has

only two nodes specific for the information of the application's users and for some pills prospectus. This aspect is visible in the photo attached next.

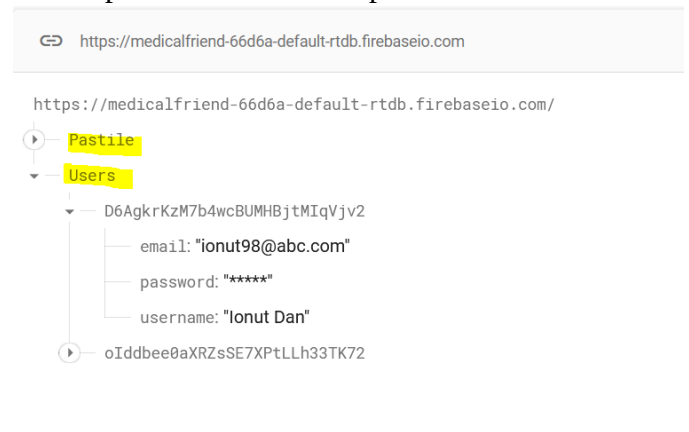


Fig. 12-Project's database structure

The graph node specific for the users informations has the following mode of separating the information specific for every user: when one person creates an account in the Authentication part of the Firebase tools will be generated an unique id. For the storage of all the informations added at the creation of the account, the java code is using the uid created to divide the information for every user.

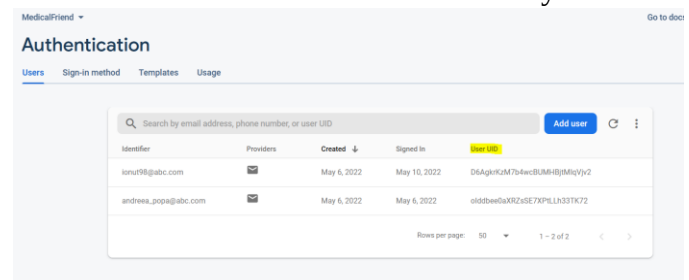


Fig. 13-Authentication type from Firebase that stores the information about every signed up user

The structure mentioned before can be seen in the *Fig. 12-Database structure*.

The second node of the application is dedicated to the implementation of the barcode scanning. With this functionality I succeed, at this point, for a limited number of drug types, to display the most important parts of a prospectus after the barcode is scanned.

As you will see in the next picture, in the database the drugs prospectus were divided by the barcode and after this separation for every instance were saved four characteristics that will be shown in the application.

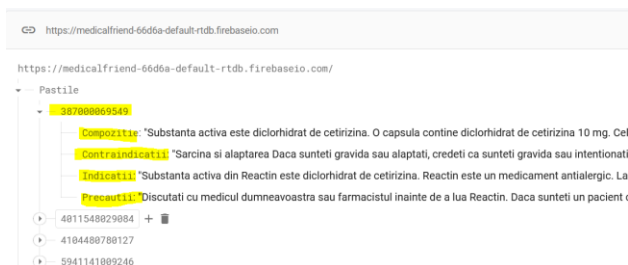


Figure 14- Barcode scanning informations in the database

Because the project assumed a lot of documentation when it comes about inserting informations, even if we talk about the section dedicated to the disease descriptions and recommendations or the section of the prospectus. For this reason when it comes about the data added for the prospectus I decided to imitate only at drugs present in Romania at this moment.

As I talk in the first chapter of this article, by developing this application I try to put together more functionalities that can only be found if we download at least two applications. I think that by putting all this together people can save time, because they will not need to open one application and then others and change between them and also will be used less of the phone memory.

When it comes about result all I can say at this point in the development cycle is that the application will be tested on multiple types of virtual phones using and android emulator and on real mobile phones. This will be done in order to see the reaction of the application when it comes to different phones displays and different versions of android systems.

V. FUTURE DIRECTIONS

As future directions for this application I am thinking about expanding the area of interest for the section dedicated to the barcode scanning, to involve prospectus for drugs specific for other countries which languages are meet in the application (e.g. Germany, France or United States).

Other implementation that I want to add to the application in the future is the possibility of GPS location of the nearest medical point in case of an emergency.

And of course the expanding of all the informational parts that are presented in the application (disease section, doctors informations- functionality that is planned to be implemented in this period of development.)

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